Disha Maheshwari

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EDUCATION

Purdue University Jan 2022 – May 2025

B.S. in Computer Engineering, GPA: 3.97/4.0 | Concentration: AI, Computer Systems, Software Engineering | Minor in Mathematics | Certificate in Applications of Data Science

Courses: Data Structures, OOP in C++, Computer Graphics, AI, Operating Systems, Signals and Systems, ASIC, NLP, Computer Security, Software Engineering, Computer Networks, Microprocessor Systems

WORK EXPERIENCE

Amazon | Software Development Engineer | Tempe, AZ Jun 2025 – Present

- Worked on feature computation platform for fraud detection: Java service handling 147.7M computations/sec and Python service handling 2.25M TPS with 3K tasks, supporting p99 latency of 20–80ms.
- Led end-to-end security migration from HTTP to HTTPS achieving 100% encryption-in-transit compliance for Tier-1 services.
- Set up service cells and infrastructure across multiple AWS regions for shadow testing and datacenter migrations.
- Performed on-call duties: incident escalation, root cause analysis, reducing operational risk, and managing high-traffic scaling.
- Enabled secure database calls by passing auth tokens from incoming requests through the request context.

Amazon | SDE Intern | Tempe, AZ May 2024 – Aug 2024

- Implemented and deployed a scalable authentication pipeline on AWS for a Java-based service; built S3 interface for periodic auth data fetching, in-memory auth refresher, and caching.

Oak Ridge National Laboratory | Research Intern | Oak Ridge, TN May 2023 – Aug 2023

- Implemented Spiking Neural Networks on Artix-7 FPGA (Basys3) in Verilog; designed UART, FIFO, and SNN modules running at 100 MHz; optimized FPGA utilization and debugged designs.
- Published a first-author IEEE paper on a low-cost, energy-efficient FPGA-based neuromorphic processor.

RESEARCH

Research Assistant | Prof. Ruqi Zhang Jun 2024 – Jun 2025

- Developed SDSI, a variational inference framework constraining posterior variances to a sparse, low-dimensional subspace for scalable Bayesian neural networks.
- Proposed a subspace selection strategy preserving high-gradient directions, enhancing accuracy and calibration on clean and corrupted datasets.
- Achieved 76.36% accuracy, 0.9655 AUROC, and 0.0014 ECE on CIFAR-100 (10% subnet), outperforming Sparse BNNs, VBLL, and Laplace Subnetworks.

[Re] BiRT: Bio-Inspired Replay in Vision Transformers for Continual Learning Nov 2023 – Jun 2024

- Conducted a reproducibility study on BiRT, a continual learning framework using Vision Transformers with constructive noises and episodic memory replay to mitigate catastrophic forgetting.
- Evaluated model performance on benchmark datasets, providing key insights into BiRT's reproducibility and practical applicability; published findings in ReScience C Journal as first author.

ADDITIONAL PROJECTS

Dual Core Cache Coherent Pipelined RISC-V Processor | SystemVerilog, Questa Sim Jan – Apr 2025

- Designed a 5-stage pipelined RISC-V CPU with variable-latency RAM, hazard handling, and branch prediction; implemented and integrated memory controller and datapath.
- Implemented split instruction/data caches with MSI coherence protocol across two cores.

Package Registry System | TypeScript, REST API, AWS Aug – Dec 2024

- Built a scalable package registry with REST APIs for module upload, versioning, and retrieval using Node.js and AWS (EC2, S3, DynamoDB).
- Implemented dependency analysis, version tracking, and cost metrics to improve system reliability and performance.
- Developed CI/CD pipeline with GitHub Actions and deployed on AWS with automated testing and continuous deployment.
- Enhanced security using HTTPS, IAM, token-based authentication, and STRIDE threat modeling.

Wisebucks.AI | Founder Jul 2023 – May 2024

- Won $6,000 in funding at JMEC startup incubator (team of 6).
- Developed an integrated platform offering financial insights, portfolio suggestions, trading simulation, and chatbot support using traditional and state-of-the-art data analysis techniques.

USB Full-Speed Bulk-Transfer Endpoint AHB-Lite SoC Module | SystemVerilog Mar – Apr 2024

- Developed a high-performance SoC module integrating a USB full-speed bulk-transfer endpoint with an AHB-Lite slave interface, data buffer, and USB RX/TX functionalities.
- Verified correctness of USB protocol handling and AHB-Lite transactions through simulation in Questa Sim.

Procedural Modelling of Terrain | OpenGL, C++, GLSL Apr – May 2023

- Used diamond-square algorithms to generate height maps; implemented GPU shaders using GLSL library.
- Built graphics pipeline including vertex shading, camera transformations, and mouse/key controls.

TECHNICAL SKILLS

Languages: Python, C, C#, C++, Java, SQL, JavaScript, TypeScript, HTML/CSS, Arduino, MATLAB, ReactJS, Verilog
ML/AI: PyTorch, Scikit-learn, NLP, LLMs, RAG, Bayesian Deep Learning, Embeddings
Cloud/DevOps: AWS (EC2, ECS, ECR, S3, DynamoDB), Docker, CI/CD, GitHub Actions, Linux, Bash
Tools: Git, VS Code, Vim, IntelliJ, Vivado, Maya, Unity, Flask, Pandas, NumPy, SciPy, OpenGL, GLSL, jQuery

PUBLICATIONS

An FPGA-based Neuromorphic Processor with All-To-All Connectivity, First Author, IEEE ICRC 2023 – doi.org/10.1109/ICRC60800.2023.10386808
[Re] Bio-Inspired Replay in Vision Transformers, First Author, ReScience C, 2025 – zenodo.org/records/14964875

AWARDS

William Hart Hayt Jr. Memorial Scholarship, Charles W. Brown Scholarship, Gold Expo Scholarship | JEE: 99.65 percentile | KVPY: AIR 205